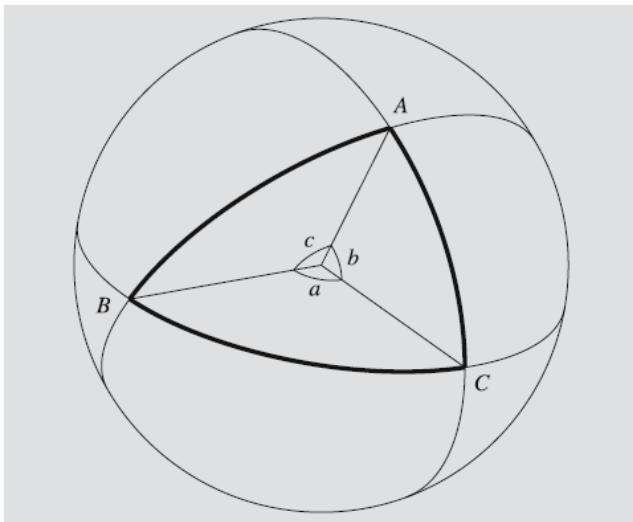


Algemeen

- Eerst een **tekening**, dan **wiskunde (formules!)**
- Eerst **wiskunde**, dan **rekenen (getallen!)**
- De **methode** is belangrijker dan het **numerieke resultaat**
- Hoeken in radialen tijdens berekening, in gepaste eenheden in resultaat ($^{\circ}, ', ''$ of $^{\text{h}}, ^{\text{m}}, ^{\text{s}}$). Afstanden in eenheden aangepast aan de omstandigheden (bvb. km of m). Gebruik gepaste nauwkeurigheid; om van radialen naar $^{\circ}, ', ''$ om te rekenen, heb je minstens 6 cijfers na de komma nodig.
- $1^{\circ} = 60' = 3600''$
- $1^{\text{h}} = 60^{\text{m}} = 3600^{\text{s}}$
- $180^{\circ} = \pi$ radialen $\rightarrow 1^{\circ} = \frac{\pi}{180}$ radialen
- $180^{\circ} = 12^{\text{h}}$ $\rightarrow 1^{\circ} = \frac{12}{180}^{\text{h}} = \frac{1}{15}^{\text{h}} = 4^{\text{m}}$



De boldriehoek ABC leeft op een sfeer met straal r .

Hoekjes en zo

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$\sin(a + b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

Booglengte

als je een lijnstuk met lengte r over een hoek c (uitgedrukt in radialen) roteert dan is de booglengte rc .

Horizontale vs. equatoriale coördinaten

$$\sin(\delta) = \mp \cos(A) \cos(a) \cos(\phi) + \sin(a) \sin(\phi)$$

$$\sin(a) = \cos(h) \cos(\delta) \cos(\phi) + \sin(\delta) \sin(\phi)$$

$$\frac{\cos(\delta)}{\sin(A)} = \frac{\cos(a)}{\sin(h)}$$

Ecliptische vs. equatoriale coördinaten

$$\sin(\delta) = \sin(\lambda) \cos(\beta) \sin(\varepsilon) + \sin(\beta) \cos(\varepsilon)$$

$$\sin(\beta) = -\sin(\alpha) \cos(\delta) \sin(\varepsilon) + \sin(\delta) \cos(\varepsilon)$$

$$\cos(\delta) \cos(\alpha) = \cos(\beta) \cos(\lambda)$$

Zonsformules

$$\sin(\delta_{\odot}) = \sin(\lambda_{\odot}) \sin(\varepsilon)$$

$$\tan(\alpha_{\odot}) = \tan(\lambda_{\odot}) \cos(\varepsilon)$$

$$\tan(\delta_{\odot}) = \sin(\alpha_{\odot}) \tan(\varepsilon)$$

Datum (benaderend)

noordelijk	Datum (2026)	$\alpha_{\odot}$	$\delta_{\odot}$	$\lambda_{\odot}$	$\beta_{\odot}$
Lente-equinox	20 maart	0h	0°	0°	0°
Zomersolsticium	21 juni	6h	+ ε	90°	0°
Herfstequinox	23 september	12h	0°	180°	0°
Wintersolsticium	21 december	18h	- ε	270°	0°

$$\lambda_{\odot}(t_2) \cong \lambda_{\odot}(t_1) + \frac{360^\circ}{365.2422 \text{ dagen}}(t_2 - t_1)$$

Zonetijd uit uurhoek van de zon

$$T = h_{\odot} + 12\text{h} - ET - \lambda + \Delta TZ$$

Tweelichamenvraagstuk

Straal ifv eccentricische anomalie

$$r = \frac{k^2/\mu}{1 + e \cos f} = \frac{a(1 - e^2)}{1 + e \cos f}$$

Energie (ellips)

$$h = \frac{1}{2}v^2 - \frac{\mu}{r} = -\frac{\mu}{2a}$$

Verband tussen energie, draaimoment en eccentriciteit

$$\mu^2(e^2 - 1) = 2k^2h$$

Derde wet van Kepler

$$P^2 = \frac{4\pi^2}{\mu} a^3$$

Vergelijking van Kepler

$$E - e \sin E = \frac{2\pi}{P}(t - \tau)$$

Date (midday UTC)	Equation of time (min)
2026-01-01	-3.56738
2026-01-02	-4.03491
2026-01-03	-4.49651
2026-01-04	-4.95169
2026-01-05	-5.39995
2026-01-06	-5.84083
2026-01-07	-6.27386
2026-01-08	-6.69859
2026-01-09	-7.11461
2026-01-10	-7.52150
2026-01-11	-7.91886
2026-01-12	-8.30632
2026-01-13	-8.68352
2026-01-14	-9.05011
2026-01-15	-9.40576
2026-01-16	-9.75018
2026-01-17	-10.08306
2026-01-18	-10.40414
2026-01-19	-10.71317
2026-01-20	-11.00990
2026-01-21	-11.29414
2026-01-22	-11.56568
2026-01-23	-11.82434
2026-01-24	-12.06997
2026-01-25	-12.30243
2026-01-26	-12.52162
2026-01-27	-12.72742
2026-01-28	-12.91976
2026-01-29	-13.09858
2026-01-30	-13.26383
2026-01-31	-13.41548
2026-02-01	-13.55353
2026-02-02	-13.67797
2026-02-03	-13.78882
2026-02-04	-13.88612

Date (midday UTC)	Equation of time (min)
2026-02-05	-13.96993
2026-02-06	-14.04030
2026-02-07	-14.09733
2026-02-08	-14.14112
2026-02-09	-14.17177
2026-02-10	-14.18941
2026-02-11	-14.19419
2026-02-12	-14.18625
2026-02-13	-14.16575
2026-02-14	-14.13288
2026-02-15	-14.08781
2026-02-16	-14.03075
2026-02-17	-13.96189
2026-02-18	-13.88145
2026-02-19	-13.78966
2026-02-20	-13.68675
2026-02-21	-13.57296
2026-02-22	-13.44854
2026-02-23	-13.31375
2026-02-24	-13.16886
2026-02-25	-13.01415
2026-02-26	-12.84990
2026-02-27	-12.67639
2026-02-28	-12.49392
2026-03-01	-12.30278
2026-03-02	-12.10328
2026-03-03	-11.89571
2026-03-04	-11.68040
2026-03-05	-11.45765
2026-03-06	-11.22778
2026-03-07	-10.99111
2026-03-08	-10.74797
2026-03-09	-10.49869
2026-03-10	-10.24359
2026-03-11	-9.98301

Date (midday UTC)	Equation of time (min)
2026-03-12	-9.71730
2026-03-13	-9.44677
2026-03-14	-9.17178
2026-03-15	-8.89266
2026-03-16	-8.60975
2026-03-17	-8.32340
2026-03-18	-8.03394
2026-03-19	-7.74171
2026-03-20	-7.44705
2026-03-21	-7.15031
2026-03-22	-6.85181
2026-03-23	-6.55190
2026-03-24	-6.25093
2026-03-25	-5.94922
2026-03-26	-5.64711
2026-03-27	-5.34495
2026-03-28	-5.04307
2026-03-29	-4.74179
2026-03-30	-4.44145
2026-03-31	-4.14237
2026-04-01	-3.84487
2026-04-02	-3.54928
2026-04-03	-3.25590
2026-04-04	-2.96506
2026-04-05	-2.67706
2026-04-06	-2.39220
2026-04-07	-2.11079
2026-04-08	-1.83313
2026-04-09	-1.55952
2026-04-10	-1.29023
2026-04-11	-1.02556
2026-04-12	-0.76579
2026-04-13	-0.51119
2026-04-14	-0.26203
2026-04-15	-0.01857

Date (midday UTC)	Equation of time (min)
2026-04-16	0.21893
2026-04-17	0.45022
2026-04-18	0.67507
2026-04-19	0.89324
2026-04-20	1.10451
2026-04-21	1.30865
2026-04-22	1.50545
2026-04-23	1.69471
2026-04-24	1.87622
2026-04-25	2.04981
2026-04-26	2.21527
2026-04-27	2.37245
2026-04-28	2.52119
2026-04-29	2.66133
2026-04-30	2.79274
2026-05-01	2.91528
2026-05-02	3.02884
2026-05-03	3.13331
2026-05-04	3.22859
2026-05-05	3.31459
2026-05-06	3.39125
2026-05-07	3.45848
2026-05-08	3.51626
2026-05-09	3.56452
2026-05-10	3.60325
2026-05-11	3.63243
2026-05-12	3.65205
2026-05-13	3.66212
2026-05-14	3.66267
2026-05-15	3.65374
2026-05-16	3.63537
2026-05-17	3.60763
2026-05-18	3.57061
2026-05-19	3.52439
2026-05-20	3.46909

Date (midday UTC)	Equation of time (min)
2026-05-21	3.40482
2026-05-22	3.33170
2026-05-23	3.24989
2026-05-24	3.15953
2026-05-25	3.06081
2026-05-26	2.95389
2026-05-27	2.83899
2026-05-28	2.71631
2026-05-29	2.58607
2026-05-30	2.44852
2026-05-31	2.30390
2026-06-01	2.15248
2026-06-02	1.99452
2026-06-03	1.83031
2026-06-04	1.66014
2026-06-05	1.48432
2026-06-06	1.30317
2026-06-07	1.11700
2026-06-08	0.92615
2026-06-09	0.73095
2026-06-10	0.53176
2026-06-11	0.32893
2026-06-12	0.12282
2026-06-13	-0.08620
2026-06-14	-0.29775
2026-06-15	-0.51145
2026-06-16	-0.72691
2026-06-17	-0.94374
2026-06-18	-1.16155
2026-06-19	-1.37994
2026-06-20	-1.59853
2026-06-21	-1.81691
2026-06-22	-2.03469
2026-06-23	-2.25149
2026-06-24	-2.46690

Date (midday UTC)	Equation of time (min)
2026-06-25	-2.68053
2026-06-26	-2.89200
2026-06-27	-3.10092
2026-06-28	-3.30690
2026-06-29	-3.50958
2026-06-30	-3.70856
2026-07-01	-3.90350
2026-07-02	-4.09402
2026-07-03	-4.27978
2026-07-04	-4.46042
2026-07-05	-4.63562
2026-07-06	-4.80504
2026-07-07	-4.96837
2026-07-08	-5.12530
2026-07-09	-5.27553
2026-07-10	-5.41877
2026-07-11	-5.55477
2026-07-12	-5.68323
2026-07-13	-5.80393
2026-07-14	-5.91659
2026-07-15	-6.02101
2026-07-16	-6.11695
2026-07-17	-6.20422
2026-07-18	-6.28261
2026-07-19	-6.35196
2026-07-20	-6.41210
2026-07-21	-6.46287
2026-07-22	-6.50415
2026-07-23	-6.53580
2026-07-24	-6.55772
2026-07-25	-6.56982
2026-07-26	-6.57200
2026-07-27	-6.56420
2026-07-28	-6.54635
2026-07-29	-6.51842

Date (midday UTC)	Equation of time (min)
2026-07-30	-6.48036
2026-07-31	-6.43216
2026-08-01	-6.37381
2026-08-02	-6.30532
2026-08-03	-6.22671
2026-08-04	-6.13800
2026-08-05	-6.03924
2026-08-06	-5.93050
2026-08-07	-5.81184
2026-08-08	-5.68333
2026-08-09	-5.54509
2026-08-10	-5.39719
2026-08-11	-5.23976
2026-08-12	-5.07292
2026-08-13	-4.89679
2026-08-14	-4.71152
2026-08-15	-4.51726
2026-08-16	-4.31416
2026-08-17	-4.10240
2026-08-18	-3.88216
2026-08-19	-3.65362
2026-08-20	-3.41700
2026-08-21	-3.17248
2026-08-22	-2.92029
2026-08-23	-2.66064
2026-08-24	-2.39377
2026-08-25	-2.11990
2026-08-26	-1.83928
2026-08-27	-1.55216
2026-08-28	-1.25878
2026-08-29	-0.95940
2026-08-30	-0.65429
2026-08-31	-0.34372
2026-09-01	-0.02797
2026-09-02	0.29269

Date (midday UTC)	Equation of time (min)
2026-09-03	0.61797
2026-09-04	0.94757
2026-09-05	1.28120
2026-09-06	1.61855
2026-09-07	1.95932
2026-09-08	2.30320
2026-09-09	2.64989
2026-09-10	2.99906
2026-09-11	3.35040
2026-09-12	3.70360
2026-09-13	4.05833
2026-09-14	4.41427
2026-09-15	4.77109
2026-09-16	5.12846
2026-09-17	5.48604
2026-09-18	5.84351
2026-09-19	6.20051
2026-09-20	6.55673
2026-09-21	6.91181
2026-09-22	7.26541
2026-09-23	7.61721
2026-09-24	7.96684
2026-09-25	8.31398
2026-09-26	8.65829
2026-09-27	8.99942
2026-09-28	9.33703
2026-09-29	9.67078
2026-09-30	10.00034
2026-10-01	10.32537
2026-10-02	10.64552
2026-10-03	10.96045
2026-10-04	11.26983
2026-10-05	11.57333
2026-10-06	11.87060
2026-10-07	12.16132

Date (midday UTC)	Equation of time (min)
2026-10-08	12.44515
2026-10-09	12.72178
2026-10-10	12.99089
2026-10-11	13.25215
2026-10-12	13.50526
2026-10-13	13.74989
2026-10-14	13.98576
2026-10-15	14.21255
2026-10-16	14.42996
2026-10-17	14.63770
2026-10-18	14.83549
2026-10-19	15.02304
2026-10-20	15.20007
2026-10-21	15.36631
2026-10-22	15.52151
2026-10-23	15.66540
2026-10-24	15.79774
2026-10-25	15.91830
2026-10-26	16.02683
2026-10-27	16.12314
2026-10-28	16.20699
2026-10-29	16.27821
2026-10-30	16.33658
2026-10-31	16.38195
2026-11-01	16.41412
2026-11-02	16.43296
2026-11-03	16.43830
2026-11-04	16.43001
2026-11-05	16.40797
2026-11-06	16.37209
2026-11-07	16.32226
2026-11-08	16.25842
2026-11-09	16.18050
2026-11-10	16.08845
2026-11-11	15.98225

Date (midday UTC)	Equation of time (min)
2026-11-12	15.86188
2026-11-13	15.72735
2026-11-14	15.57867
2026-11-15	15.41588
2026-11-16	15.23902
2026-11-17	15.04818
2026-11-18	14.84343
2026-11-19	14.62489
2026-11-20	14.39267
2026-11-21	14.14692
2026-11-22	13.88781
2026-11-23	13.61551
2026-11-24	13.33023
2026-11-25	13.03219
2026-11-26	12.72163
2026-11-27	12.39882
2026-11-28	12.06403
2026-11-29	11.71755
2026-11-30	11.35971
2026-12-01	10.99082
2026-12-02	10.61124
2026-12-03	10.22133
2026-12-04	9.82148
2026-12-05	9.41208
2026-12-06	8.99357
2026-12-07	8.56637
2026-12-08	8.13093
2026-12-09	7.68772
2026-12-10	7.23721
2026-12-11	6.77990
2026-12-12	6.31629
2026-12-13	5.84689
2026-12-14	5.37222
2026-12-15	4.89282
2026-12-16	4.40923

Date (midday UTC)	Equation of time (min)
2026-12-17	3.92200
2026-12-18	3.43167
2026-12-19	2.93882
2026-12-20	2.44399
2026-12-21	1.94777
2026-12-22	1.45072
2026-12-23	0.95341
2026-12-24	0.45642
2026-12-25	-0.03968
2026-12-26	-0.53432
2026-12-27	-1.02693
2026-12-28	-1.51695
2026-12-29	-2.00383
2026-12-30	-2.48701
2026-12-31	-2.96597